NEUROBOLT PROJECT · RESEARCH GUIDE

EEG, BOLD and the arousal axis

What a year of reproducing and dissecting an EEG-to-fMRI transformer actually taught us, what holds up on a new dataset, a working prototype built on it, and what to do next, written to be learned from and explained to others.

0.381 vs 0.336
Ridge regression beats the NeuroBOLT transformer at predicting a new person's BOLD (mean R, 22 people)

~60%
of what EEG predicts is the brain-wide (global) signal, an arousal read-out

0.227 = 0.228
On 28 new people at another site, the frozen model matches one trained there: no site penalty

1 · The one-page summary
2 · The question and why it matters
3 · Key concepts, in depth
4 · What we found on the VU data
5 · Does it hold on a new dataset?
6 · A prototype: the EEG arousal monitor
7 · The larger research questions
8 · Resources needed
9 · Working as a human researcher alongside AI
10 · What you are learning, and how to say it
References

Seamus Taylor · Monash University · October 2026

Data: VU resting-state EEG-fMRI (22 people, 29 scans; NeuroBOLT, NeurIPS 2024) and OpenNeuro ds006040 (28 people; Cha et al., Sci Data 2026). Computation: Monash MASSIVE M3. Code and full reports: the project repository.

1

The one-page summary

The question. Can scalp EEG, cheap and portable, tell us what deep brain activity measured by fMRI is doing in a person the model has never seen? NeuroBOLT (NeurIPS 2024) said yes, using a pretrained EEG transformer. We reproduced it, then took it apart.

Five things to know

1. **A simple linear model does better than the transformer.** Ridge regression on EEG power in 1 Hz bins over the last 24 s predicts a new person's BOLD at mean R = 0.381 across all 22 people, against 0.336 for NeuroBOLT. NeuroBOLT's pretrained transformer branch contributes nothing; a linear model reproduces its output at R = 0.91.
2. **What EEG predicts is mostly arousal.** The best-predicted regions are those that follow the brain-wide global signal. Removing that signal cuts accuracy from 0.34 to 0.14. A one-number vigilance index carries a third of the explained variance, and heart rate alone predicts BOLD at R = 0.22.
3. **It is real physiology, not an artefact.** 28 of 29 scans beat their own null; one person's prediction scored against another person's BOLD gives R = 0.00. The model rediscovers the known alpha–BOLD coupling (negative in cortex about 9 s later, positive in the thalamus about 3 s later).
4. **More model capacity does not help at this data size.** MLPs, splines, interactions, learned filters and per-person calibration all tie or trail ridge. A compact interpretable model keeps 96% of ridge's skill with 0.2% of its parameters.
5. **It transfers to a new site.** Frozen and applied to 28 people scanned elsewhere (ds006040), the model predicts eyes-closed BOLD as well as a model trained there (R = 0.227 against 0.228). The physiology replicates, and ridge still beats NeuroBOLT. Eyes-open rest is a real shift (0.14).

What we built

An EEG arousal monitor (section 6): it estimates, every fMRI volume and from EEG alone, the arousal-driven brain-wide BOLD signal and a vigilance trace, flags low-vigilance periods and scores how much arousal moved in a run. On the new dataset it tracks the measured global BOLD at r = 0.26 (eyes closed), runs live at about 3 ms per fMRI volume, and its arousal-fluctuation score tracks the fMRI-measured global-signal swing across runs (ρ = +0.50). Its drowsiness flags are not yet validated.

What this means

Resting-state EEG tells us where a person sits on the arousal axis, delayed by the haemodynamic response, and linear models already read all of it. The ceiling is set by how much arousal varies in a recording and by the diversity of the people and sites in the training data, not by model architecture. The open scientific question, and the publishable one, is the *region-specific remainder*: what EEG predicts about a brain region once the brain-wide signal is removed.

> DECISIONS THIS GUIDE ASKS THE TEAM TO MAKE
>
> 1. Adopt region-specific (global-signal-removed) accuracy and the mismatched-person null as the benchmark for any future EEG-to-fMRI model.
> 2. Fund the external-data track: two to three more open datasets processed through one shared pipeline (section 8).
> 3. Decide whether to develop the arousal monitor towards a real-time pilot in the scanner (section 6).

2

The question and why it matters

Two ways of watching the brain. EEG records electrical activity from the scalp with millisecond timing, costs little and fits on a head. fMRI measures the blood-oxygen-level-dependent (BOLD) signal everywhere in the brain, including deep structures such as the thalamus, but needs a scanner, is slow (one image every 2 s) and sees neural activity only through the sluggish blood response. Recording both at once lets us learn the mapping between them.

Why predicting fMRI from EEG would matter. If EEG alone could tell us what deep brain regions are doing, we could monitor them outside the scanner: in sleep labs, at the bedside, during neurofeedback, or over days rather than minutes. The test that matters for those uses is *cross-subject*: train on some people, predict BOLD in someone new.

What NeuroBOLT claimed. Li et al. (NeurIPS 2024) combined a pretrained EEG foundation model (LaBraM) with a multi-scale spectral branch and reported mean R = 0.473 for seven brain regions in five held-out people. Our project asked whether that holds up, and if so, what the model has learned.

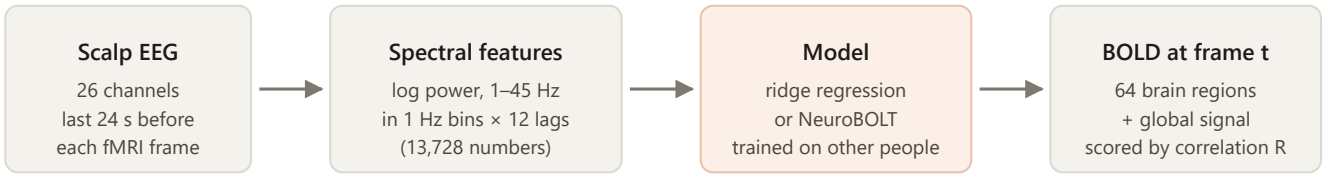


Figure 1. The prediction problem. The model sees only EEG from before the frame, so it is causal and could, in principle, run live.

The project in three stages

Stage	What we did	Outcome
Reproduction	Retrained NeuroBOLT for 7 regions and all 29 scans on M3; scored each held-out scan separately	Cross-subject result reproduced (0.46 vs 0.473); within-scan result did not (0.38 vs 0.53)
Hypothesis tests	33 NeuroBOLT variants, 21 band-power baselines, 5-fold cross-validation over all 22 people	Ridge wins for 19–20 of 22 people; the paper's split is the easiest of five folds
Interpretability	Nulls, whole-brain maps, physiology controls, model dissection, nonlinear tests, an interpretable time-series model, personalisation	Section 4: the signal is arousal, read linearly through the haemodynamic delay
External validation and prototype	A new site (ds006040, 28 people) and an application	Sections 5 and 6

3

Key concepts, in depth

Ten ideas carry every result in this guide. Each comes with why it matters here and the mistake it guards against.

3.1 The haemodynamic response: BOLD is late and smooth

Neurons do not light up fMRI directly. Activity triggers a local blood-flow increase that peaks about 5–6 s later and settles over 15–20 s: the haemodynamic response function (HRF). So BOLD at time t reflects neural activity from roughly t − 4 to t − 12 s. That is why the models look at 24 s of EEG history, and why the most useful EEG is 8–12 s before the frame. The HRF also varies between regions and people by a second or two; a textbook HRF shape fitted to these data loses 40% of the accuracy, so the models learn their own delays.

3.2 EEG band power and the vigilance index

Raw EEG is summarised by its power at each frequency. In eyes-closed rest, **alpha** (8–13 Hz) is strong when a person is relaxed but awake; as they drift towards sleep, alpha fades and slower **theta** (4–8 Hz) and **delta** (1–4 Hz) rise. The ratio of alpha to delta+theta is therefore a standard EEG vigilance index. For twenty years, studies have shown that alpha power goes with *lower* BOLD in visual and parietal cortex a few seconds later, and with *higher* BOLD in the thalamus (Goldman et al. 2002; Laufs et al. 2003; de Munck et al. 2007).

3.3 Encoding models

An encoding model predicts a brain measurement from features of the input, here BOLD from EEG features. It is a predictive association, not a causal claim: a good prediction means the EEG carries information about the BOLD, not that one causes the other. Both may follow a third process, which in this project turns out to be arousal.

3.4 Cross-validation by person, and what *n* really is

Every number here is scored on people the model never saw. The data are split into five groups of people; each group is held out once while the model trains on the rest, and settings such as the penalty are chosen on separate validation people. Two consequences matter. First, a 20-minute scan has 560 frames, but they are strongly autocorrelated, and what we want to generalise across is people: the real sample size is 22, not 16,000. Second, cross-validation with small samples has wide error bars (around ±10% even at 100 samples; Varoquaux 2018), and a single split can mislead. NeuroBOLT's own split turned out to be the easiest of the five (0.46–0.52 against 0.19–0.40 elsewhere).

3.5 R, R^2 and paired comparisons

R is the correlation between predicted and measured BOLD over a held-out scan. $\mathbf{R^2}$ is the fraction of variance explained, so R = 0.38 means about 14%; halving R quarters the variance explained, which is why "half the R" can be "a third of the variance". To compare two models fairly, compute each person's difference and put a confidence interval on those 22 differences ("paired" comparison). Differences under about 0.02 R are usually within noise here.

3.6 Null models: what would chance look like?

Slow, smooth time series correlate by chance much more than independent numbers do; over a 10-minute scan, chance |R| of 0.1–0.2 is common. Two nulls answer two different worries. A **circular-shift null** (slide the measured BOLD in time against the prediction) asks whether the scan's own autocorrelation could produce the R. A **mismatched-person null** (score one person's prediction against another person's BOLD) asks whether

something shared by every scan, such as drowsiness that builds the same way in everyone, inflates it. The second is the decisive check, and the literature rarely runs it.

3.7 The global signal and arousal

Averaging BOLD over the whole brain gives the **global signal**. It is not noise: a large part of it follows arousal, rising as vigilance falls, and it is coupled to heart rate, breathing, pupil size and EEG (Chang et al. 2016; Bolt et al. 2025). Arousal changes BOLD most in sensory cortex and in the opposite direction in parts of the thalamus and cingulate. **Global-signal regression (GSR)** removes, per scan, the part of each region that follows the global signal, leaving what is region-specific.

3.8 Ridge regression and why simple models win with few people

Ridge regression is linear regression with a penalty that shrinks all weights towards zero, trading a little bias for a large cut in variance. With 13,728 features and 13–17 training people, that trade dominates. Learning theory and benchmarks agree on the general picture:

- **Bias–variance.** A flexible model fits the training people better but varies more between training sets. With few independent units, variance wins unless the extra flexibility matches real structure.
- **Interpolation and double descent** (Belkin et al. 2019; Hastie et al. 2022; Bartlett et al. 2020) explain why very large models can still generalise, but those results rest on specific conditions about the data's covariance and on large samples; in their own analyses, tuned ridge remains hard to beat in the small-sample linear regime.
- **Scaling laws** (Kaplan et al. 2020; Hoffmann et al. 2022): deep networks improve as a power law in data, so a 22-person dataset sits at the very start of the curve. In brain imaging, linear models matched deep ones up to about 10,000 people (Schulz et al. 2020).
- **Inductive bias.** When the true mapping is close to linear in the right features, a model that assumes that structure needs far less data. Log band power with lags is that structure here.

3.9 Reading a model: weights, patterns and importance

With correlated features, raw weights mislead: a weight can be large because it cancels noise in a neighbouring feature. The safe objects (Haufe et al. 2014) are **patterns** (how each feature co-varies with the model's prediction), **group refits** (what is lost when a block of features is removed) and **matched perturbations** (scramble one part of the input and measure the damage). **Surrogates** help too: if a simple model can reproduce a complex model's output, the complex model has, in effect, learned the simple one.

3.10 Transfer and domain shift

A model trained at one site meets different amplifiers, caps, artefact-cleaning pipelines, scanners and people at another. Some shift is in feature *levels* (the whole spectrum is louder or quieter), which standardising the new site's EEG with its own statistics can absorb without any fMRI. Some is in the *relationship* itself, for example eyes open versus eyes closed, where alpha behaves differently. External validation measures how much survives.

4

What we found on the VU data

4.1 Ridge beats NeuroBOLT, for nearly everyone

5-fold CV, all 22 people (mean R, 7 regions)	R	People where ridge wins
Ridge, 1 Hz bins × 24 s	0.381	–
NeuroBOLT, shared model, 24 s window	0.336	19 of 22
NeuroBOLT, shared model, 16 s (paper setting)	0.323	20 of 22

Of 33 NeuroBOLT changes, only a longer EEG window helped. Its pretrained LaBraM transformer branch never learned (R = 0.02 alone); the spectral branch did all the work.

4.2 The prediction is real

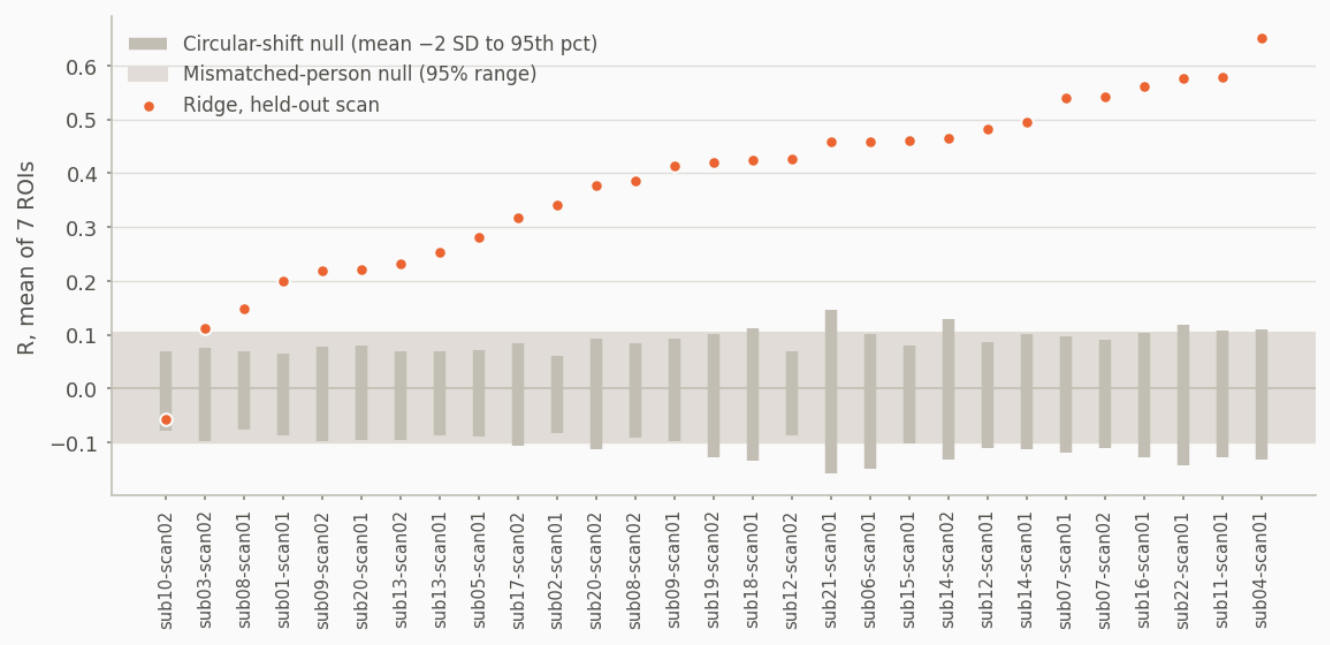


Figure 2. Each orange dot is one held-out scan. Grey bars: that scan's circular-shift null; grey band: mismatched-person null. 28 of 29 scans clear their null; the mismatched-person null is centred on zero (−0.003).

4.3 It is mostly the brain-wide arousal signal

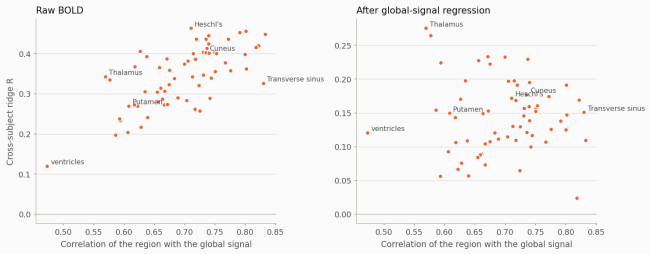


Figure 3. Left: a region is predictable to the extent it follows the global signal (ρ = +0.65 over 64 regions). Right: after GSR the relation vanishes.

Sensory and motor cortex are predicted best (R ≈ 0.43), control and default-mode regions worst (≈ 0.27), which is the spatial signature of arousal. Removing the global signal per scan cuts mean R from 0.34 to 0.14. What survives is far above chance, and concentrated where arousal moves BOLD *against* cortex: the thalamus (0.28), cerebellum (0.26) and mid-anterior cingulate (0.23).

Five channel-averaged band powers keep 93% of the skill; scalp topography adds nothing. A one-number vigilance index explains about a third of what the full model explains.

4.4 The model reads known physiology through the haemodynamic delay

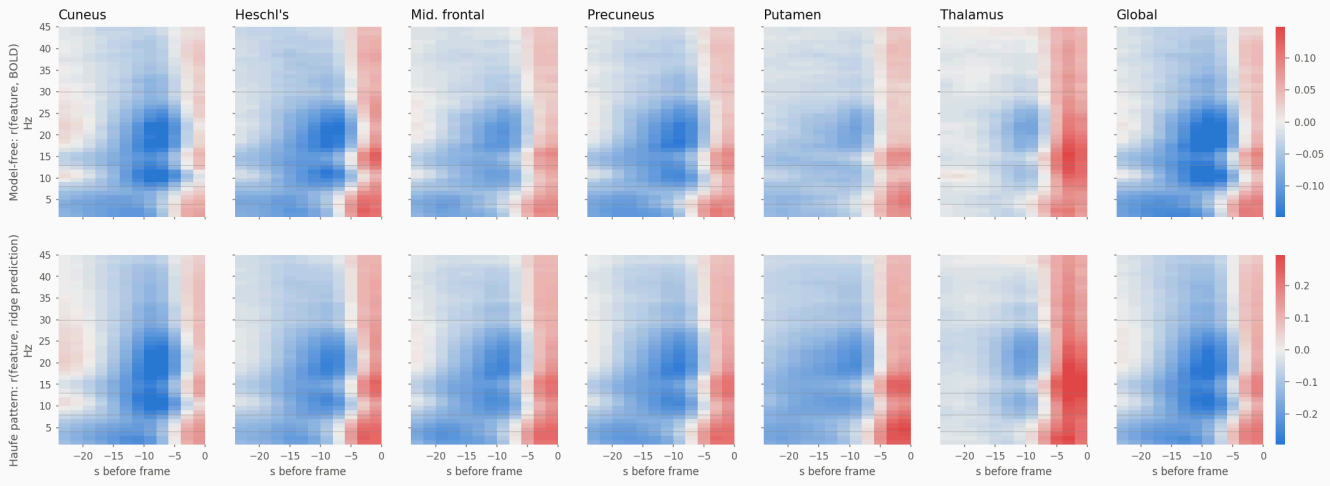


Figure 4. Top: model-free correlation of EEG power (frequency on the y axis, seconds before the frame on the x axis) with BOLD, averaged over 29 scans. Bottom: the same for ridge's predictions. Blue: more power goes with lower BOLD; red: higher.

- **Cortex:** alpha and beta power 8–12 s before a frame predict lower BOLD ($r \approx -0.25$, $t \approx -12$ over scans), the classic inverse coupling seen through the HRF.
- **Thalamus:** the sign flips and the delay shortens; alpha and beta predict higher BOLD with a 3 s lag ($r \approx +0.20$ to $+0.24$), as within-subject studies reported, here holding across people.
- **The last 4 s:** broadband power just before a frame predicts higher BOLD everywhere. This survives in the model's patterns, so it is real; it looks like arousal transients that move EEG and BOLD together.

4.5 NeuroBOLT learned the same linear map

A ridge model trained to imitate NeuroBOLT's own predictions reproduces them at $R = 0.91$ on held-out scans and predicts BOLD better than NeuroBOLT does (0.491 against 0.465): the non-linear part of its output is mostly noise. Scrambling parts of the EEG moves both models alike, with the last 4 s and the 8–12 s lobe mattering most.

4.6 Brain or body?

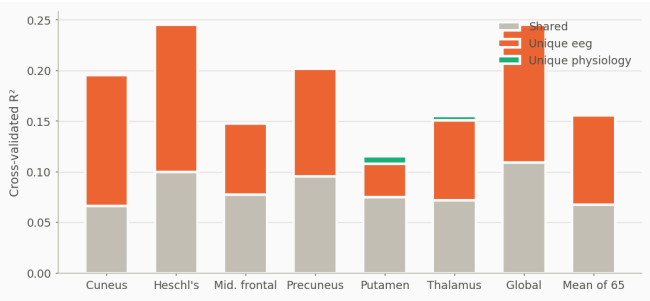


Figure 5. Cross-validated variance explained, split into the part shared by EEG and body signals (grey), unique to EEG (orange) and unique to physiology (green, almost nothing).

Arousal also changes heart rate, eye movements, muscle tone and head motion. Using only measures EEG cannot contaminate (heart rate, eye channels below 4 Hz, muscle above 30 Hz), the body predicts BOLD at $R = 0.26$ with no EEG at all, and heart rate alone at 0.22.

But everything the body predicts, the EEG already predicts: they share about 44% of EEG's explained variance, and physiology adds almost nothing unique (R^2 +0.001). Head-motion sensors add nothing beyond EEG. The body signals are co-markers of the same state, not separate routes that EEG misses.

4.7 Capacity does not help; structure is low-dimensional

Readout of the same features	Protocol	R (7 regions)	Δ vs ridge (95% CI)
MLP, 60 band powers	5-fold	0.361	+0.006 (−0.012, +0.024)
MLP, 60 band powers	leave one person out	0.375	+0.010 (−0.007, +0.026)
MLP, 1,560-feature cube	leave one person out	0.378	+0.005 (−0.014, +0.024)
Additive splines / vigilance × band	5-fold	0.347	−0.008 / −0.008
Learned frequency bands (end to end)	5-fold	0.355	−0.026
Person-specific delay from half the scan	calibrated	0.406	+0.006 (−0.019, +0.031)
Interpretable model: 5 shared components, 2,040 parameters	5-fold, 65 targets	0.329	−0.015 vs 0.344 for 892,320 ridge weights

A 2026 preprint (MEL, same data) reported a large MLP gain; under its own leave-one-person-out protocol we find +0.01, not significant. The interpretable "Hammerstein" model, a few shared scalp-spectral components each with its own regional delay kernel, keeps 96% of ridge's skill, and one component alone reaches 88%: the predictable part of resting BOLD is low-dimensional.

5

Does it hold on a new dataset?

Everything above came from one site. We took the model trained on the VU people, froze it, and applied it to 28 people scanned at another site, on another scanner, with another EEG cap and artefact-cleaning pipeline.

> **THE NEW DATASET: OPENNEURO DS006040**
>
> Cha et al., *Scientific Data* 2026, Sungkyunkwan University; 28 people; 3T Siemens Prisma; TR 2.0 s; 64-channel BrainCap MR EEG (it contains all 26 channels the model uses); eyes-closed and eyes-open rest, about 6.3 minutes each, with the scanner on. The authors removed MR artefacts from the EEG and preprocessed the fMRI with fMRIPrep. We rebuilt the same features and targets as for VU and dropped frames touching stretches of missing EEG (9 of 56 runs had some).

5.1 The model transfers with no site penalty

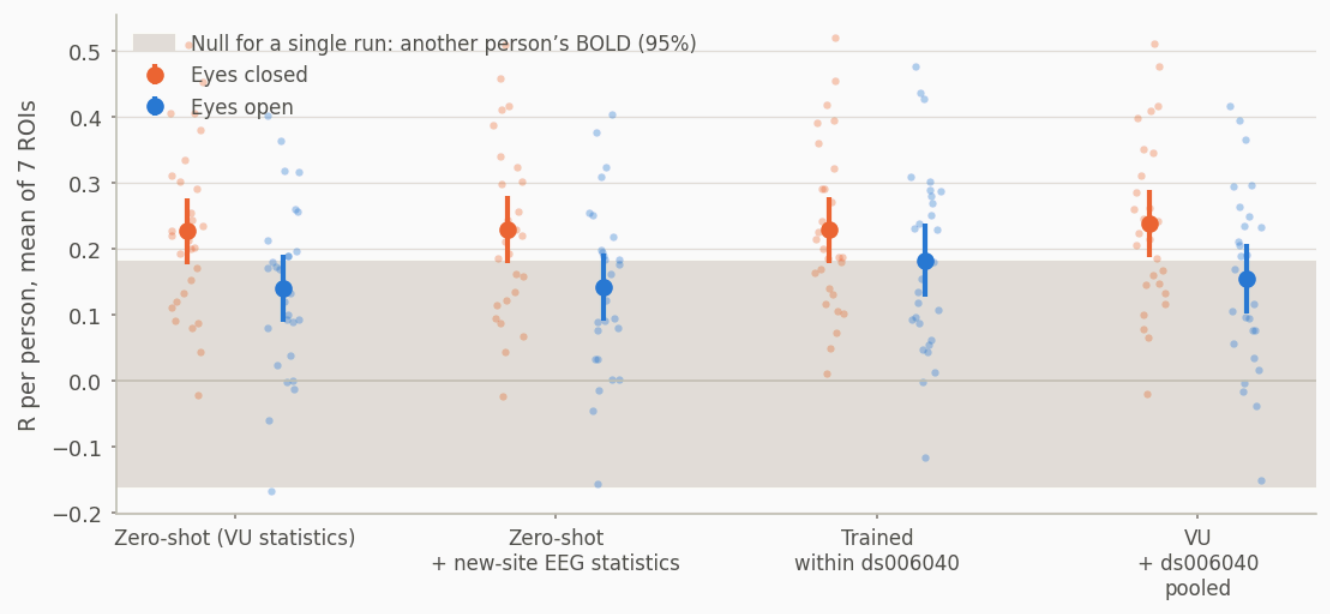


Figure 6. Each dot is one person (mean of 7 regions); large dots and bars are means with 95% CIs. Grey: the null for a single run (another person's BOLD); the group means sit far above its mean of 0.00.

Full ridge model, 7 regions	Eyes closed	Eyes open	After GSR (eyes closed)
Frozen VU model (zero-shot)	0.227	0.140	0.106
Trained within ds006040 (5-fold)	0.228	0.182	0.119
VU + ds006040 pooled	0.238	0.155	0.132
NeuroBOLT, authors' checkpoints (zero-shot)	0.180	0.140	–

- **No site penalty.** The imported model is exactly as good as one trained on the new site's own people (paired Δ 0.000, 95% CI −0.018 to +0.018); pooling both sites adds a non-significant +0.010.
- **Lower level, same ceiling.** R is lower than on VU (0.23 against 0.38) for both the imported and the local model, so the cause is the data: 6-minute runs of mostly alert people carry less arousal fluctuation than 20-minute eyes-closed scans.
- **Region-specific signal transfers too** (0.106 zero-shot, 0.132 pooled after GSR), which supports question 1 in section 7 as a benchmark that generalises.

- **Eye state matters.** Eyes-open rest drops the frozen model to 0.14, while a model trained on eyes-open data reaches 0.18: the vigilance index barely works with eyes open (0.045), so other spectral features carry the signal.
- **Ridge still beats NeuroBOLT** on the new eyes-closed data by 0.05 (95% CI 0.03 to 0.07); with eyes open they tie.

5.2 The physiology replicates

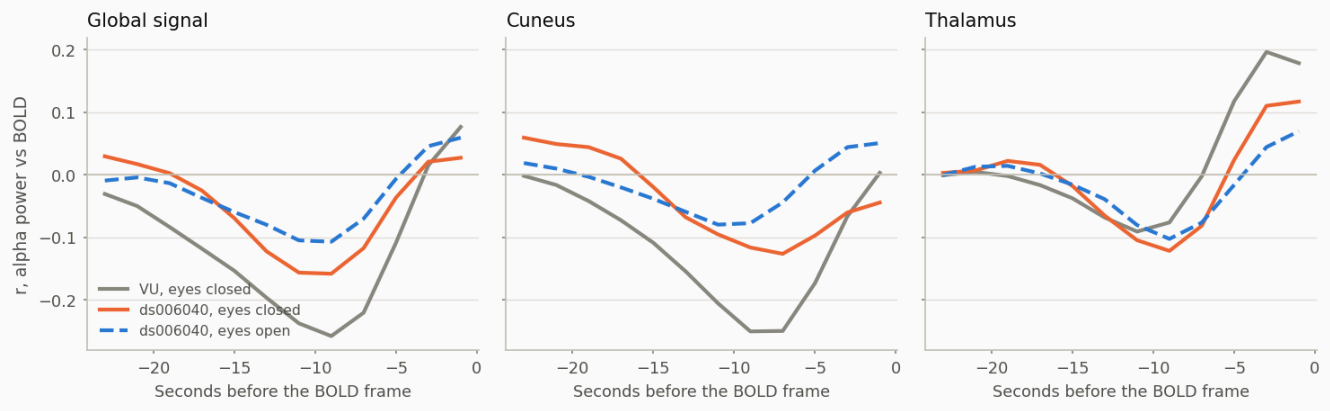


Figure 7. Correlation of alpha power with BOLD by lag. Grey: VU; orange: ds006040 eyes closed; blue dashed: eyes open.

The inverse alpha coupling of cortex and the global signal, peaking about 9 s before the frame, replicates at about 60% of its VU strength. The thalamus keeps its positive short-lag lobe (beta +0.16, alpha +0.11 at 1–3 s) but shows a stronger negative lobe at 9 s than on VU, so the thalamic sign flip is the least robust signature and needs a larger sample.

5.3 What this tells us

- **The finding is general, not a quirk of one dataset.** A model that encodes "arousal, seen through the haemodynamic delay" works on new people, a new scanner and a new cap without retraining.
- **Accuracy is set by the recording more than by the model.** The same 0.23 ceiling for imported and local models, and lower accuracy on short, alert runs, point to arousal variance as the limiting quantity.
- **Two open questions move to the front:** eyes-open data need their own calibration, and the thalamic signature needs a larger, pooled sample to settle.

6

A prototype: the EEG arousal monitor

The research says EEG is an excellent, causal read-out of the arousal-driven part of brain activity. That is useful on its own, so we built a tool around it.

6.1 What it does and who would use it

The **EEG arousal monitor** takes scalp EEG and, once per fMRI volume, estimates the brain-wide BOLD fluctuation that arousal drives, plus an EEG vigilance trace. Because each estimate uses only the preceding 24 s of EEG, it can run live during a scan.

- **Scan-time quality control.** Resting-state fMRI is confounded by participants drifting towards sleep: about half lose wakefulness within 10 minutes (Tagliazucchi and Laufs 2014). The monitor shows this as it happens, instead of after preprocessing.
- **An arousal covariate** for analyses that need to model or report vigilance.
- **EEG-only settings**, such as sleep labs or bedside monitoring: an fMRI-calibrated arousal index without a scanner.

6.2 How it works

Pick 26 standard channels, resample to 200 Hz, compute log power in 1 Hz bins over twelve 2 s slices of the last 24 s, standardise within the recording (or with running statistics when live), and apply the frozen ridge model trained on VU. The code is a small Python package (*app/arousal_monitor*) with offline and streaming modes and a command-line interface; it refuses frames that touch missing EEG.

6.3 Validation on 56 runs it never saw

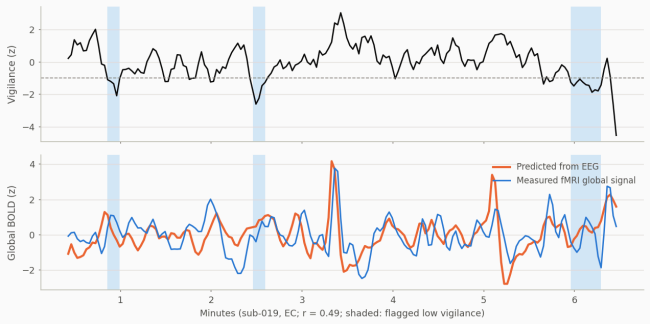


Figure 8. The best-tracked eyes-closed run (r = 0.49); typical runs track at r ≈ 0.25.

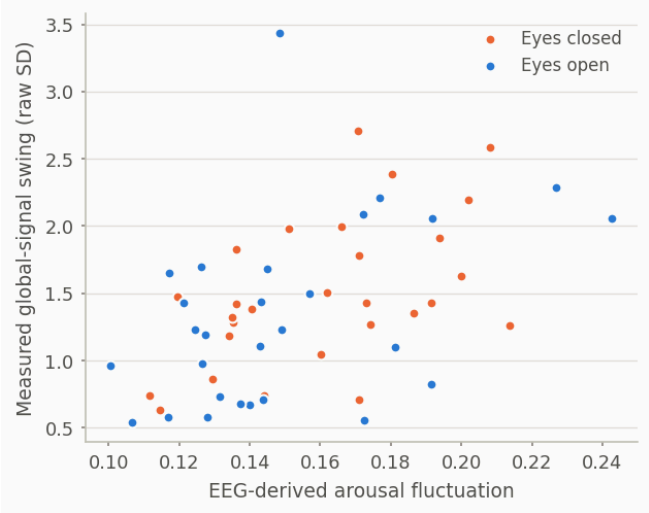


Figure 9. Across runs, the monitor's arousal-fluctuation score tracks the global-signal swing measured by fMRI (ρ = +0.50).

Measure	Eyes closed	Eyes open
Tracks measured global BOLD (r, offline)	0.26 (0.20–0.31)	0.17 (0.11–0.24)
Live streaming mode	0.26 (0.21–0.32)	0.16 (0.10–0.22)
Computation per fMRI volume		3 ms on one CPU core
QC score vs measured global-signal swing (ρ, 56 runs)		+0.50 ($p < 0.001$)
QC score vs how predictable the run is		+0.42 ($p = 0.001$)

Validated: the per-volume estimate, the live mode, and the run-level QC score. **Not yet validated:** the low-vigilance flags, which need eye tracking or sleep scoring to check; the raw vigilance index's fluctuation did not track the fMRI measures. The QC score also tracks head motion ($\rho = +0.32$), consistent with arousal and movement rising together, but a motion-matched test is needed. The interactive demo replays all 56 runs: claude.ai/artifact/M3MtScwGda3GP4d2e27dUU.

> **NEXT STEPS FOR THE PROTOTYPE**
>
> 1. Validate the flags on a dataset with eye tracking or sleep staging (NATVIEW, the ds003768 sleep runs).
> 2. Connect to a live EEG stream (BrainVision RDA or Lab Streaming Layer) and pilot it in the scanner.
> 3. Calibrate an eyes-open variant and test it on ds007216, which has self-reported arousal probes.

7

The larger research questions

Each question below is answerable with existing open data and a clear test; the first two are the strongest bets for a publication.

Q1. What does EEG know about a brain region beyond brain state?

After removing the global signal, EEG still predicts region-specific BOLD (mean R 0.14), strongest in the thalamus, cerebellum and cingulate. No published EEG-to-fMRI study reports this number. **Test:** make region-specific accuracy, scored against the mismatched-person null, the primary benchmark; ask whether any model beats ridge there. A positive answer would make two decades of within-subject EEG-fMRI findings testable across people; a negative one would show resting-state EEG-to-fMRI models are vigilance decoders.

Q2. What sets the ceiling: the model, the data, or the person?

Accuracy ranged from 0.23 to 0.52 between groups of people, while the learning curve flattened after about nine training people and capacity added nothing. People whose arousal fluctuates are predictable. **Test:** train and test across sites and across arousal ranges (sleep recordings), and model accuracy as a function of each run's arousal variance. If sleepier, longer recordings raise the ceiling, the bottleneck is the signal, not the model.

Q3. Is the thalamic coupling a cross-person marker of arousal regulation?

The thalamus is the one region where coupling flips sign and arrives earlier. **Test:** estimate thalamic and cortical delay kernels per person in larger samples, check test–retest reliability (NATVIEW, ds007216 have repeat sessions), and relate them to behaviour (sustained attention in ds006040's gradCPT task).

Q4. Can an EEG-derived arousal signal replace or complement physiological recordings in fMRI analysis?

EEG explained everything the body signals explained and more. **Test:** on a dataset with respiration and pulse (ds005795), compare EEG-derived, physiology-derived and fMRI-derived (global signal) arousal regressors for denoising and for connectivity estimates.

Q5. When does nonlinearity start to pay?

Theory predicts deep models need much more data per pattern. **Test:** pool about 150 people from open datasets and add hours-long sleep recordings; re-run the ridge-versus-MLP comparison with paired per-person tests. The answer decides whether foundation-model approaches are worth their cost in this field.

Q6. Can the arousal monitor improve real studies?

Test: a prospective pilot in which the monitor runs live during resting scans, with blinded comparison of flagged periods against eye-tracking and post-scan sleepiness reports.

8

Resources needed

Resource	What and why	Estimate
Data	Open simultaneous EEG-fMRI with rest: ds003768 (33, same TR and amplifier as VU, sleep runs), ds005795 (34, eyes closed, motion loops, respiration), NATVIEW (22 × 2 sessions), ds007216 (24 × 2), ds005127 (16, overnight sleep). About 178 people in total, all CC0/CC BY.	0.5–1 TB download; free; data-use terms are open
Preprocessing	One shared pipeline for every site: MR artefact correction of EEG where only raw data exist (ds003768), fMRIPrep, DiFuMo-64 extraction. Site differences must not come from different pipelines.	~3–6 core-hours per person for fMRIPrep; ~1 week of engineering
Compute	Everything in this guide ran on M3: ridge and analyses on CPU nodes, NeuroBOLT and MLPs on single GPUs (4-GPU quota). Pooled 150-person models stay within this.	< 2,000 CPU-hours and < 300 GPU-hours per study; 2 TB scratch
People and skills	One researcher with signal processing and statistics (core); input from an EEG-fMRI physiologist (arousal, thalamus), an MRI physicist for pipelines, and a statistician for pre-registration and paired designs.	1 FTE for 6–12 months, plus advisory time
Real-time pilot	Scanner time with simultaneous EEG, a live data stream (BrainVision RDA or LSL), ethics approval, 10–15 participants.	Ethics 2–3 months; ~20 scanner hours
Time	External validation on two more sets (2 months); region-specific benchmark paper (3–4 months); pooled-data nonlinearity study (4–6 months); real-time pilot (6–9 months).	About 12 months for the programme

RISKS WORTH NAMING

- **Pipeline confounds.** If sites are preprocessed differently, "site differences" may be pipeline differences. Mitigation: one pipeline for every site; section 5 shows ridge needed no site adjustment, but check this for every new site.
- **Small effects, small samples.** Model differences of 0.01–0.02 R need dozens of people to resolve. Mitigation: pre-registered thresholds, paired tests, and pooling.
- **Over-reading arousal as region-specific function.** Always report raw and GSR results side by side.

9

Working as a human researcher alongside AI

Much of the computation, coding and literature search in this project was done with an AI assistant directing jobs on M3. That changes what the human researcher's job is. This section draws on what current research says about AI capabilities and failure modes, and on this project's own experience, to separate what to delegate from what to own.

9.1 What the evidence says about AI as a research tool (2025–2026)

Finding	What it means for us
Task length is growing fast. The length of software tasks AI agents complete at 50% reliability has doubled roughly every 7 months since 2019 (Kwa et al. 2025, METR).	Long, well-specified pipelines (fetch data, build features, run cross-validation) are now delegable. Reliability at a given length is still 50%, not 99%: check outputs.
Feeling faster is not being faster. In a randomised trial, experienced developers took 19% longer with early-2025 AI tools, while believing they were 20% faster (Becker et al. 2025, METR).	Measure your own productivity on real tasks; do not trust the impression of speed.
Replication is harder than description. On PaperBench, the best agent replicated 21% of ICML papers' results, about half the human baseline (Starace et al. 2025); agents describe methods well but lose points executing them.	Ask the AI to run things, then verify numbers against an independent check (we added nulls, assertions and cross-checks against cached results throughout).
Citations need checking. Papers with at least one fabricated reference rose from about 1 in 2,800 (2023) to 1 in 277 (early 2026) in a 2.5-million-paper biomedical audit (Topaz et al., Lancet 2026); 2.6% of 2025 papers at three CS conferences had a likely hallucinated citation.	Every reference in this guide was found by search, but treat machine-extracted numbers as unverified until you have read the source. Our literature review flags preprints and machine-extracted figures for that reason.
Models agree too readily. Leading models changed answers to match a user's pushback in 58% of tested cases, sometimes from right to wrong (Fanous et al. 2025, SycEval).	Do not ask "is this right?" with your preferred answer in the question. Ask for the strongest case against it, and for pre-registered thresholds.
Real discoveries, mostly in reach of search. AI systems resolved several open Erdős problems in 2025–26, but a DeepMind study screening 700 "open" problems concluded that many were open through obscurity rather than difficulty, and several were already solved in the literature (Feng et al. 2026). Google's AI co-scientist produced drug-repurposing hypotheses that worked in vitro, after experts chose which to test (Gottweis et al. 2025).	AI is strong at breadth: search, recombination, exhaustive checking. Humans still choose which problems matter and which hypotheses are worth testing.

9.2 Where human judgement was decisive in this project

- **Shared-reference trap.** The eye, muscle and heart channels share the EEG reference, so their power partly *is* EEG. A naive analysis said "eye channels predict BOLD almost as well as EEG"; the correct control used only measures EEG cannot contaminate. Knowing the recording hardware mattered more than the model.
- **Choosing the null.** Statistical significance per scan was easy; the question that mattered, whether all scans share a time-locked drift, needed a different null. The mismatched-person null answered it.
- **Not believing a striking preprint.** The MEL preprint's MLP gain would have redirected the project. Re-running it under its own protocol showed it does not replicate on our features.
- **Reading the short-lag lobe.** An odd positive term near zero lag could have been dismissed as a weight artefact; checking it in model-free correlations showed it is real and probably arousal transients.

9.3 Specific findings a human researcher should look into

1. **Inspect raw EEG at high-leverage moments.** Look at the 4 s before frames with the largest predicted BOLD swings: are they arousals, movements, eye opening, or residual artefact?
2. **Check pipeline equivalence.** The VU data were cleaned by the authors with tools we did not see; ds006040 with FMRIB and median-template BCG removal. Confirm with the authors what was done, and how ROI series were denoised.
3. **Ask the NeuroBOLT authors** how the within-scan epochs were chosen; picking the best test epoch reproduces their 0.53, validation choice gives 0.38.
4. **Read the MEL preprint in full** and contact its authors about their preprocessing and protocol before citing the discrepancy.
5. **Interpret the thalamic result physiologically** with a domain expert: which nuclei (pulvinar couples the other way), and whether it reflects arousal regulation or vascular timing.
6. **Decide thresholds before results.** For each next experiment, write the expected effect and the falsifier first (the literature review's scorecard is a template).

A WORKING PROTOCOL FOR RESEARCH WITH AI

- **Delegate breadth, own depth:** AI for literature sweeps, scaffolding code, running many controlled variants; you for problem choice, causal interpretation and deciding what counts as surprising.
- **Pre-register, then run:** state expected effects and falsifiers before the AI sees results; it keeps both of you honest.
- **Demand controls by default:** nulls, paired comparisons, a simple baseline and an independent replication for any headline.
- **Verify the chain, not the prose:** spot-check data alignment, counts and units; read primary sources for any number you will repeat.
- **Keep a decision log** (the project's memory notes and reports do this) so that a result can be traced to the code and data that produced it.

10

What you are learning, and how to say it

10.1 What you are actually learning

Area	The transferable lesson
Neurophysiology	EEG and BOLD are linked mainly through arousal and neurovascular delay; the thalamus behaves differently from cortex; the global signal is physiology, not just noise.
Statistics	The unit of inference is the person; autocorrelated signals need their own nulls; paired designs and confidence intervals over people; small samples have wide error bars.
Machine learning	Strong baselines first; capacity must earn its place with data; surrogates and matched perturbations reveal what a model really uses; a striking claim is a hypothesis until replicated under its own protocol.
Research practice	Separate reproduction from extension; pre-register thresholds; validate externally; document so others can rerun everything (M3 scripts, memory notes, reports).
Working with AI	Specify tasks precisely, check numbers independently, and keep scientific judgement human.

10.2 Three sentences for stakeholders

> 1. "A simple, transparent model predicts a new person's fMRI from their EEG better than the published deep-learning model, and we can show exactly what it uses."
> 2. "What EEG mostly tells us about the resting brain is the person's level of alertness, seen through the slow blood response; that holds on an independent dataset: a model trained at one site works unchanged on 28 people scanned elsewhere."
> 3. "We have turned that into a working prototype that tracks arousal-driven brain activity from EEG alone, and the next step is to test what EEG knows about specific brain regions beyond alertness."

10.3 What not to claim

- Not "EEG can replace fMRI": it predicts mainly the brain-wide arousal component; region-specific accuracy is about 0.14.
- Not "deep learning doesn't work for EEG": it does not help *at this data size and for this target*.
- Not "the authors were wrong": their cross-subject number reproduces on their split; their split was unusually easy.
- Not "EEG causes BOLD changes": these are predictive associations with a shared cause.

10.4 Questions you will be asked

Question	Short answer
Is R = 0.38 good?	It is about 14% of variance, well above chance (null ≈ 0.00 ± 0.05) and above the published model; typical within-person EEG-fMRI correlations in the literature are 0.2–0.3.
Why not just use more data with the transformer?	That is Q5; theory and our learning curve suggest diversity of people and arousal range matter more than more of the same scans.
Could it be head motion or the heart?	Those signals are shared with EEG and add nothing unique; motion loops add +0.001.
Will it work on other scanners?	Section 5: for eyes-closed rest, yes: on 28 people at another site the frozen model matches one trained there. Eyes-open rest needs its own calibration.
What is the product?	An arousal monitor for fMRI sessions and EEG-only settings (section 6); a research tool, not a clinical device.

10.5 Glossary

BOLD: the fMRI signal, reflecting blood oxygenation. **HRF**: the haemodynamic response function, the delayed blood response to neural activity. **Alpha/theta/delta**: EEG rhythms at 8–13, 4–8 and 1–4 Hz. **Global signal**: BOLD averaged over the brain. **GSR**: removing it from each region. **Ridge**: penalised linear regression. **Cross-validation**: repeatedly training on some people and testing on others. **Null**: the distribution of a score when there is no real relation. **Haufe pattern**: how input features co-vary with a model's output, the readable counterpart of its weights. **DiFuMo-64**: a 64-region brain atlas. **Domain shift**: differences between training and deployment data.

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Full project reports: *NeuroBOLT reproduction results*, *NeuroBOLT hypothesis tests*, *NeuroBOLT interpretability*, *NeuroBOLT external validation*, and the 117-citation literature review *EEG BOLD interpretability and directions*, all in the repository's reports folder.